

Guandi Wang

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EDUCATION

- **KTH Royal Institute of Technology** Stockholm, Sweden
M.Sc. in ICT Innovation: Autonomous Systems and Intelligent Robots Aug. 2025 – Present
- Selected Coursework: Machine Learning, Distributed Artificial Intelligence, Robotics, Hybrid and Embedd Control.
- **Xi'an University of Architecture and Technology** Shaanxi, China
B.Sc. in Building Electrical and Intelligence Sep. 2021 – July 2025
- GPA: 83.5/100 (Top 10%) | **Technical Focus:** Control Theory, Power Engineering, Building Environment, Embedded Systems.

RESEARCH EXPERIENCE

Spatio-Temporal Weather Analysis and Forecast (Research Project) Sep. 2025 – Present
Developing a high-resolution meteorological benchmark utilizing ERA5 reanalysis data, focusing on wind speed forecasting in Southeast China's coastal zones.

Designed a data pipeline to process large-scale grid meteorological data, addressing complex spatio-temporal dependencies suitable for data-driven modeling.

Evaluating various Deep Learning baselines on High-Performance Computing (HPC) clusters using Slurm, optimizing for computational efficiency.

Deep learning network designing.

Robust Multimodal Object Detection via Spatial Feature Rectification(Research Project)Jun. 2025 – Dec. 2025
Proposed a Scale-Modality Decoupled Interaction module to disentangle scale-invariant patterns from modality-specific features (RGB & Thermal).

Developed a feature-space augmentation strategy using attention-guided convex combinations to synthesize hard examples, improving model robustness in adverse conditions.

Conducted extensive experiments on standard benchmarks , demonstrating superior performance in handling sensor misalignment.

Autonomous Mobile Robot Navigation System (Coursework) Sep. 2025 – Dec. 2025
Architected a navigation stack for TurtleBot3 using **ROS2 (Foxy)** and Python. Migrated control logic from Finite State Machines to **Behavior Trees** for enhanced reactivity.

Integrated AMCL for robust localization and tuned PID controllers for smooth trajectory tracking.

Photovoltaic Panel Defect Detection Algorithm | *Bachelor Thesis* Jan. 2025 – Jun. 2025
Constructed a dual-modal (Visible-Infrared) dataset for PV defects. Implemented image registration using Harris corners and RANSAC algorithms.

Optimized a YOLOv8-based fusion model, designing ablation studies to validate the effectiveness of different fusion strategies (Early vs. Late Fusion).

TECHNICAL SKILLS

Languages: Python , C/C++, MATLAB, LaTeX

Frameworks & Libraries: PyTorch, ROS2 (Foxy/Humble), OpenCV, Scikit-learn, Slurm (HPC)

Tools & Platforms: Linux (Ubuntu), Git, Docker, Gazebo

SELECTED AWARDS

Grand Prize, Undergraduate Graduation Project Competition, Shaanxi Automation Association 2025
Outstanding Graduate, Xi'an University of Architecture and Technology 2025
Academic Scholarship, Class II, Awarded for 3 consecutive years 2021 – 2023